Geometric Stability Theory Reading Group

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1 Notation and Conventions

Symbols:

$\mathfrak{C}$ denotes the monster model.

$S^n(A)$ is the space of complete types in n-variables over A.

Unless otherwise specified, $S(A) = S^1(A)$.

We write $\phi(x) \leq \psi(x)$ if $\mathfrak{C} \vDash \forall x.(\phi(x) \to \psi(x))$.

We follow the convention of [TZ12] for the monster model, treating it as a proper class sized "model" which contains all set-sized models as elementary substructures. By **large** we mean something proper class sized, whereas by **small** we mean something set sized.

All our theories are complete. Later, we will generally restrict to stable theories, though we will state when this is the case.

2 Week 1 - Imaginaries and Stability

In this week, we cover pages 7-17 of [Pil96], with a diversion at the end to discuss the properties of Cantor-Bendixson rank in more detail. We have mostly omitted proofs in favour of a broader discussion of important definitions, though most proofs can be found in the book.

2.1 Elimination of Imaginaries

Given the monster model $\mathfrak{C}$ of a theory T in a single-sorted language L, its **imaginary elements** are the equivalence classes of $\emptyset$-definable equivalence relations on definable classes. Much of stability theory can be simplified if we are allowed to treat imaginary elements of the monster as if they were bona fide elements. This is called **elimination of imaginaries**. We describe here a uniform method of generating a theory with elimination of imaginaries from T that has the same stability properties. The concession we must make is to move to an infinitely sorted language.

Definition 2.1. Given a $\emptyset$-definable equivalence relation E on $\mathfrak{C}^n$ and an E-class X, a **code** for X is a tuple $\overline{d} \in \mathfrak{C}$ such that, for every automorphism $\sigma \in \mathrm{Aut}(\mathfrak{C})$:

$$\sigma(\overline{d}) = d \text{ if and only if } \sigma(X) = X.$$

Remark 2.2. A code may also be referred to as a **name** or a **canonical parameter**.

Definition 2.3. A theory T is said to have **elimination of imaginaries** if for every $\emptyset$-definable equivalence relation E and E-class X, there is a code for X in $\mathfrak{C}$.

We now define our extended language L^{eq}, which is dependent on T. This language will have a sort for every $\emptyset$-definable equivalence relation and projection maps from the base sort onto a representative for its equivalence class in the new sorts.

Definition 2.4. Given an L-theory T,

- Let $ER(T) = \{E \subset \mathfrak{C}^{2n} \mid E \text{ is a definable equivalence relation on } n-\text{tuples}\}$.
- Let the sorts of L^{eq} be S_E for each $E \in ER(T)$. We highlight a **base sort** $S_=$.
- L^{eq} includes all the function and relation symbols of L, but we now specify them to be on $S_=$.
- For each $E \subset \mathfrak{C}^{2n}$ in $ER(T)$, we have a projection map $\pi_E \colon (S_=)^n \to S_E$.

Remark 2.5. We obtain a natural embedding of L into L^{eq} where every L-formula $\phi(x_1, \dots, x_n)$ becomes an L^{eq} formula by converting all bound and free variables to variables of the sort $S_=$.

We can now extend $\mathfrak{C}$ and T.

Definition 2.6. We define the structure $\mathfrak{C}^{\text{eq}}$ as follows:

- For each $E \in ER(T)$, S_E is interpreted in $\mathfrak{C}$ as the class of equivalence classes of E in $\mathfrak{C}$.
- In particular, we have $(S_=)^{\mathfrak{C}^{\text{eq}}} = \mathfrak{C}$.
- All the function and relation symbols from L will be interpreted the same as in $\mathfrak{C}$ on the base sort.
- For each $E \in ER(T)$, we interpret π_E as sending tuples $\overline{a} \in \mathfrak{C}$ to their E-equivalence class in $(S_E)^{\mathfrak{C}^{\text{eq}}}$.

Definition 2.7. The theory T^{eq} is the L^{eq}-theory given by:

- All the axioms of T appropriately adjusted to the base sort.
- For each $E \in ER(T)$, an axiom stating that π_E is surjective. That is, $\forall y \in S_E . \exists x \in S_= . (\pi_E(x) = y)$.
- For each $E \in ER(T)$, an axiom stating that $\pi_E(\overline{a}) = \pi_E(\overline{b})$ if and only if $E(\overline{a}, \overline{b})$. That is, $\forall \overline{x}, \overline{y} \in S_= . [(\pi_E(\overline{x}) = \pi_E(\overline{y})) \leftrightarrow E(\overline{x}, \overline{y})]$.

Remark 2.8. The structure $\mathfrak{C}^{\text{eq}}$ is a model of T^{eq}. Indeed, it is the monster model for T^{eq}.

We note that the same process can indeed be done for any model M of T to obtain a model M^{eq} of T^{eq}. Indeed, we obtain the following important fact:

Lemma 2.9. *Every model of T^{eq} is of the form M^{eq} for M a model of T.*

We also obtain results telling us that T^{eq} is a *conservative* extension of T, introducing essentially no new logical facts.

Lemma 2.10. *T^{eq} is complete.*

Lemma 2.11. *For $E_1, \dots, E_k \in ER(T)$ and $\phi(x_1, \dots, x_n) \in L^{\text{eq}}$ where each x_i is of the sort S_{E_i}, there is an L-formula $\psi(y_1, \dots, y_n)$ such that*

$$T^{\text{eq}} \vDash \forall \overline{y}_1, \dots \overline{y}_n \in S_{=}.[\phi(\overline{y}_1, \dots, \overline{y}_n) \leftrightarrow \phi(f_{E_1}(\overline{y}_1), \dots, f_{E_n}(\overline{y}_n))].$$

Remark 2.12. In particular, for every L^{eq}-formula with variables only from $S_{=}$, there is some L-formula equivalent to it. That is, every definable subclass of $\mathfrak{C}^n$ in T^{eq} is already definable in T.

Proposition 2.13. *T^{eq} has elimination of imaginaries.*

Proof. Let $E \in ER(T)$ and X be an E-class. Then, there is an element in $(S_E)^{\mathfrak{C}^{\text{eq}}}$ corresponding to X. This element is a code for X. □

We thus see that T^{eq} and its corresponding monster model are the correct place for us to work in when we need elimination of imaginaries. From hereon, we shall simply assume we are working in T^{eq} whenever we discuss a theory T.

A notational convenience that this construction also allows is that, for any given tuple of elements in $\mathfrak{C}^{\text{eq}}$, there is a single element that this tuple maps to in the sort for equivalence relation of pointwise equality of tuples of the appropriate length. Thus, we will often conflate elements and tuples, though we shall be more specific if it is necessary for clarity.

2.2 Stability

We now turn to the notion of stability, both its global version for a whole theory and local version at a specific formula.

Definition 2.14. A formula $\delta(x, y)$ is said to have the **order property** if there are sequences $(a_i)_{i<\omega}, (b_i)_{i<\omega}$ such that $\vDash \delta(a_i, b_j)$ if and only if $i \leq j$.

Definition 2.15. A formula without the order property is called **NOP** or **stable**. A theory is called **stable** if every formula is stable.

Lemma 2.16 ([Pil96] Lemma 1.2.1)**.** *1. The negation, conjunction and disjunction of stable formulae are stable.*

2. *If $\psi(y, x) = \phi(x, y)$ then $\phi(x, y)$ is stable if and only if $\psi(y, x)$ is stable.*
3. *If T is stable then every formula in L^{eq} is stable.*
4. *A formula $\phi(x, y)$ is stable if and only if there is some $n < \omega$ there are no finite sequences $(a_i)_{i=1}^n$ and $(b_i)_{i=1}^n$ such that $\vDash \delta(a_i, b_j)$ if and only if $i \leq j$.*

We also introduce the notion of types localised at a given formula and their definitions.

Definition 2.17. Let $\delta(x, y)$ be an L-formula.

- For $a \in \mathfrak{C}$ an element, we call the $L(a)$-formula $\delta(x, a)$ an **instance** of δ.
- A δ-formula is a boolean combination of instances of δ and negations of instances of δ.
- A complete δ-type over A is a maximal consistent collection of δ-formulae with parameters from A.
- The collection of complete δ-types over A is written $S_\delta(A)$.

Remark 2.18. Given a type $p \in S(A)$, we write $p \restriction \delta$ for the δ-type given by all δ-formulae in p.

Definition 2.19. Given a model M and a δ-type $p(x) \in S_\delta(M)$, a δ-**definition** of p is a formula $\psi(y) \in L(M)$ such that:

$$\text{For all } m \in M, \delta(x,m) \in p(x) \text{ if and only if } M \vDash \psi(m).$$

We shall also state some other equivalent characterisations, though these will not all be used.

Proposition 2.20 ([Pil96] Remark 1.2.3)**.** *Let $\delta(x,y)$ be an L-formula. Then, the following are equivalent:*

(i) *$\delta(x,y)$ is stable.*

(ii) *For every model M and every δ-type $p(x) \in S_\delta(M)$, $p(x)$ has a δ-definition.*

(iii) *For every cardinal $\lambda \geq |T|$ and parameter set A with $|A| \leq \lambda$, we have $|S_\delta(A)| \leq \lambda$.*

(iv) *There exists a cardinal $\lambda \geq |T|$ such that for every parameter set A with $|A| \leq \lambda$, we have $|S_\delta(A)| \leq \lambda$.*

Once we have stability, we can in fact specify the form of the defining formulae for our types.

Proposition 2.21 ([Pil96] Lemma 1.2.2)**.** *Let $\delta(x,y)$ be a stable formula, M be a model and $p(x) \in S_\delta(M)$. Then, $p(x)$ has a δ-definition $\psi(y)$ which is a positive boolean combination of instances of δ with parameters from M.*

2.3 R_δ-rank

We now introduce a useful tool for the study of stable formula: an ordinal-valued rank function called the δ-rank, written R_δ, which is defined on complete types. This will initially be defined on **global types**, which are proper class sized types over the monster model. For those familiar with Cantor-Bendixson rank from topology, the δ-rank of a type is simply its Cantor-Bendixson rank in the stone space $S_\delta(\mathfrak{C})$.

Definition 2.22. We recursively stratify the space $S_\delta(\mathfrak{C})$ as follows:

1. $S_{\delta,0}(\mathfrak{C}) = S_\delta(\mathfrak{C})$.
2. For an ordinal α, $I_{\delta,\alpha}(\mathfrak{C})$ is the set of isolated points in $S_{\delta,\alpha}(\mathfrak{C})$. More precisely, this is $\{p \in S_{\delta,\alpha} \mid \exists \phi(x) \in L(\mathfrak{C})$ such that p is the unique type in $S_{\delta,\alpha}$ containing $\phi\}$.
3. For an ordinal α $S_{\delta,\alpha+1}(\mathfrak{C}) = S_{\delta,\alpha}(\mathfrak{C}) \setminus I_{\delta,\alpha}(\mathfrak{C})$.
4. For a limit ordinal γ, $S_{\delta,\gamma} = \bigcap_{\beta<\gamma} S_{\delta,\beta}(\mathfrak{C})$.

Using this stratification, we have:

- The δ-rank of $p(x) \in S_\delta(\mathfrak{C})$ is given by $R_\delta(p) = \sup\{\alpha \mid p(x) \in S_{\delta,\alpha}(\mathfrak{C})\}$. If $p(x) \in S_{\delta,\alpha}(\mathfrak{C})$ for all ordinals α, we write $R_\delta(p) = \infty$. We will say that the rank is **unbounded** in this case.
- Given a parameter set A and a type $p(x) \in S_\delta(A)$ we say $R_\delta(A) = \max\{R_\delta(q) \mid q(x) \in S(\mathfrak{C})$ and $q \supseteq p\}$.
- Given a δ-formula $\phi(x) \in L(A)$, we say $R_\delta(\phi) = \max\{R_\delta(q) \mid q(x) \in S(A)$ such that $\phi(x) \in q\}$.
- Given an element a, we write $R_\delta(a/A) = R_\delta(\mathrm{tp}(a/A) \upharpoonright \delta)$.

We can now state some properties of this rank.

Proposition 2.23 ([Pil96], Lemma 1.3.3)**.** *Let $\phi(x)$ and $\psi(x)$ be δ-formulae with bounded rank. Then the following hold:*

1. *For every automorphism $\sigma \in \mathrm{Aut}(\mathfrak{C})$, $R_\delta(\sigma(\phi)) = R_\delta(\phi)$.*
2. *If $\phi \leq \psi$ then $R_\delta(\phi) \leq R_\delta(\psi)$.*
3. *$R_\delta(\phi \wedge \psi) \leq \min\{R_\delta(\phi), R_\delta(\psi)\}$.*
4. *$R_\delta(\phi \vee \psi) = \max\{R_\delta(\phi), R_\delta(\psi)\}$.*
5. *There is some type $p(x) \in S(\mathfrak{C})$ containing $\phi(x)$ such that $R_\delta(\phi) = R_\delta(p)$. This property of a rank is called **continuity**.*

Inside a stable formula, we obtain good behaviour of this rank.

Proposition 2.24 ([Pil96] Lemma 1.2.6)**.** *Let $\delta(x, y)$ be a stable formula. Then, $R_\delta(p) < \infty$ for all $p(x) \in S_\delta(\mathfrak{C})$.*

References

[Pil96] Anand Pillay. *Geometric Stability Theory*. 1996 (cit. on pp. 4, 7–10).

[TZ12] Katrin Tent and Martin Ziegler. *A Course in Model Theory*. Cambridge University Press, 2012 (cit. on p. 3).